Article

AERIS: Environment-Aware Hierarchical Routing for Reliable Wireless Sensor Networks under Realistic Channel Conditions

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Abstract: This paper presents AERIS, an environment-aware hierarchical routing protocol for wireless sensor networks, evaluated using packet delivery ratio (PDR). In a four-environment 100-node matrix (30 seeds per environment), AERIS attains the highest mean PDR among LEACH, PEGASIS, HEED, and TEEN. In a stricter large-scale matrix (100–1000 nodes) with balanced sampling (3200 replicates per environment-node-protocol cell) and explicit MAC-collision plus multi-hop relay modeling, AERIS remains rank-1 in three environments, while PEGASIS is strongest in indoor_office at high scale. Matched tx-power tests (5 vs 15 dBm, n=600 per cell) show strong environment-dependent responses. Matched patch-control tests (n=1000 vs 1000) show consistent absolute PDR degradation under strict physics settings, indicating calibration requirements for deployment. NS-3 is used for cross-platform trend validation of AERIS versus LEACH. Overall, the evidence positions AERIS as a reliability-oriented option for harsh channels with explicit environment-scoped operating boundaries.

Keywords: wireless sensor networks; reliable routing; hierarchical protocol; packet delivery ratio; scalability; reproducible evaluation

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1. Introduction

Wireless sensor network (WSN) routing is classically optimized for energy reduction, but practical deployments also require stable delivery under heterogeneous channels [1–3]. This creates a multi-objective tension among reliability, hop-based latency, and energy use. Existing protocols represent distinct operating points: LEACH emphasizes clustered forwarding [4], PEGASIS emphasizes chain-based aggregation [5], HEED emphasizes residual-energy-aware head selection [6], and TEEN emphasizes event-triggered reporting [7]. Prior channel studies also show that low-power links can be asymmetric and environment-sensitive under realistic propagation conditions [8–10].

AERIS is designed as a lightweight hierarchical protocol that prioritizes reliability under realistic channel variability. The design objective is not universal dominance; rather, it is robust performance across multiple channel environments while preserving deployability on resource-constrained nodes.

Contributions

This manuscript makes three contributions:

- a modular routing design (CAS switching, Skeleton backbone, Gateway reinforcement) with explicit environment-scoped operation boundaries and auditable control variables;
- publication-tier evidence matrices that separate core reliability reporting (100-node + primary large-scale matrix) from stress/sensitivity calibration blocks;
- cross-platform NS-3 trend validation for AERIS versus LEACH under aligned environment taxonomy.

The manuscript focuses on reproducible conclusions under explicit settings (metric, environment taxonomy, baseline set, and sample-size design).

2. Related Work

Classical WSN protocols (LEACH, PEGASIS, HEED, TEEN) provide foundational routing mechanisms but show different failure modes under harsh channels and larger scales [3,5,6,11]. Chain-based aggregation can reduce per-hop transmission effort but increases end-to-end hop count. Cluster-based approaches can reduce hop count but may become sensitive to uplink quality and head placement.

Recent adaptive or learning-based methods improve specific settings but often require heavier runtime or training overhead [12–17]. Recent Sensors/IoT studies report gains from context-aware or hybrid routing under constrained networks [18–24]. Additional recent studies show that routing quality is strongly coupled to link asymmetry, deployment context, and adaptive control-policy design [25–29]. These results motivate explicit environmental adaptation in protocol design while keeping inference and control logic lightweight. In this manuscript, the focus is a rule-based protocol with explicit reproducibility constraints and auditable experiment metadata. The baseline set (LEACH, PEGASIS, HEED, TEEN) is selected to span representative WSN routing paradigms (cluster-head rotation, chain forwarding, residual-energy clustering, and threshold-triggered event reporting) under a comparable low-overhead control budget.

3. System Model and Protocol

3.1. Metric Definition

The primary metric is

$$\mathrm{PDR}_{\mathrm{expected}} = \frac{\text{bs\_delivered}}{\text{source\_packets\_expected}}. \tag{1}$$

All publication conclusions in this draft are based on this metric.

3.2. AERIS Architecture

AERIS combines three components:

1. Context-Adaptive Switching (CAS) for mode control,
2. Skeleton backbone for hierarchical forwarding,
3. Gateway coordination for CH-to-BS reinforcement.

Gateway candidate scoring follows a weighted rule

$$\mathrm{Score}_i = \alpha E_i + \beta C_i + \gamma L_i \tag{2}$$

where E_i is residual energy, C_i is centrality, and L_i is link quality. CAS mode selection is modeled as

$$m^* = \arg \max_{m \in \{\text{direct,chain,twohop}\}} \left(w_{1,m}\hat{q}_m + w_{2,m}\hat{e}_m + w_{3,m}\hat{u}_m + b_m \right), \tag{3}$$

where $\hat{q}_m$, $\hat{e}_m$, and $\hat{u}_m$ are normalized link-quality, energy, and uncertainty features for mode m. The CAS coefficients $\{w_{1,m}, w_{2,m}, w_{3,m}, b_m\}$ are fixed per environment family from a pre-publication calibration step and then held constant across all publication-tier runs; they are not tuned on the evaluation replicates reported in this manuscript.

3.3. Reliability-Gain Intuition

The key design intuition is path diversity under uncertain links. Let p_d, p_g, and p_s denote effective successful-delivery probabilities for direct CH uplink, Gateway-assisted uplink, and Skeleton-assisted uplink under the same round state. If these alternatives are available and independently attempted within protocol constraints, the combined success probability can be written as

$$P_{\text{succ}} = 1 - (1 - p_d)(1 - p_g)(1 - p_s), \tag{4}$$

which is non-decreasing with each component probability under an independence approximation. We use Eq. (4) as design intuition rather than as an exact stochastic model because forwarding paths are partially correlated in practice. CAS then chooses the lower-cost intra-cluster collection mode before this uplink stage, so routing overhead is bounded while uplink diversity is preserved.

3.4. Design Rationale and Decision Logic

The protocol design follows three engineering objectives that are directly linked to known failure modes of classical WSN routing.

- **Reliability under harsh channels**: when direct CH-to-BS links become unstable, Gateway support and Skeleton routing create additional forwarding opportunities instead of relying on a single uplink path.
- **Low online overhead**: CAS is a fixed-coefficient rule model with three candidate modes; no online training or iterative optimization is introduced at runtime.
- **Auditable behavior**: all mode choices are deterministic under the same input state, and publication-tier runs fix the coefficient set per environment family.

At each round, AERIS first performs CH election and clustering, then applies CAS mode selection for intra-cluster forwarding, and finally performs CH-to-BS forwarding with optional Gateway/Skeleton reinforcement. This ordering separates local data-collection decisions from long-range forwarding decisions and avoids cross-stage coupling in the control loop.

3.5. Round Pseudocode and Complexity

Round-level pseudocode

1. Compute CH scores and elect CH set.
2. Associate non-CH nodes to CHs.
3. For each cluster, evaluate CAS features $(\hat{q}, \hat{e}, \hat{u})$ and select one mode from {direct, chain, twohop}.
4. Aggregate data at CH.
5. If Gateway is enabled, rank gateway candidates with Eq. (2); if Skeleton is enabled, build/update backbone links.
6. Forward CH traffic to BS and update metrics.

Let N be the node count and H be the CH count in one round. CH election and clustering are approximately $O(NH)$ in the current implementation, CAS selection is $O(1)$ per cluster decision, and Gateway ranking is $O(H \log H)$ when sorting candidates. Skeleton update and relay-path maintenance add an additional graph-update term that scales with active candidate links E_a. Therefore, the dominant per-round cost is approximated as

$$T_{\text{round}} = O(NH + H \log H + E_a), \tag{5}$$

where $H \ll N$ in the tested regimes. This complexity expression is an implementation-level bound for the current simulator, not a formal optimality claim.

3.6. Round-Level Operation

Each simulation round follows a fixed sequence: (i) cluster-head election and clustering, (ii) intra-cluster delivery with CAS mode selection, (iii) CH-to-BS forwarding with optional Gateway/Skeleton support, and (iv) metric logging. This ordering is held constant across protocols to preserve comparability in the reported matrices. Figure 1 summarizes this execution and evidence pipeline.

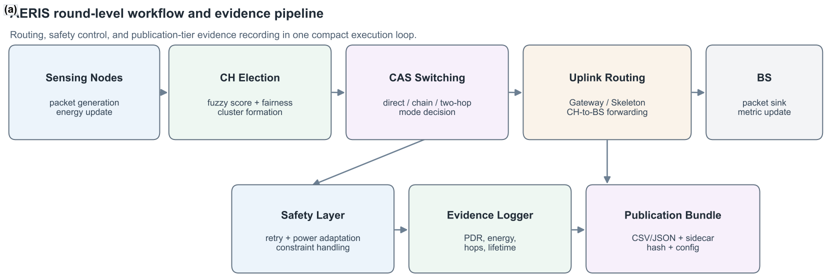

Figure 1. AERIS workflow from round execution to publication-tier evidence output. The diagram separates routing decisions, safety control, and metric/provenance recording to make reproducibility boundaries explicit.

4. Experimental Setup

4.1. Publication-Tier Settings

- Simulator: Python implementation with logged provenance.
- Channel environments: indoor_office, indoor_factory, outdoor_urban, outdoor_suburban.
- Default node count: 100.
- Seeds for multi-environment and ablation: 30 independent seeds.
- Seeds for scalability: primary large-scale matrix with n=3200 per environment-node-protocol cell (100–1000 nodes, 4 environments; strict-physics flags enabled).
- Baselines: LEACH, PEGASIS, HEED, TEEN.

4.2. Evidence Scope

The analysis is grounded in publication-tier multi-environment comparison, ablation, scalability, hop-based latency, energy-lifetime statistics, and NS-3 trend validation. Cross-regime numerical pooling is avoided.

4.3. Fairness Controls and Simulator Credibility

The simulator is custom Python code, so fairness controls are explicitly enforced in publication-tier runs:

- identical tx-power configuration across protocols in the same matrix,
- identical environment taxonomy and node-scale grid per protocol,
- deterministic seed schedules per regime,
- no reliability override (`force_ctp_reliable=False`).

To reduce implementation-bias risk, the primary large-scale matrix removes previously identified baseline suppressors and reruns the four-environment scalability matrix under explicit collision and relay settings. NS-3 is then used as an independent trend-level check for AERIS versus LEACH under aligned environment definitions. NS-3 is therefore used as external directional validation rather than as a one-to-one numerical replacement of the Python matrix.

4.4. Simulator Regimes and Physics Assumptions

This study reports two regime tiers for scalability interpretation. The primary tier is the large-scale matrix (explicit `-mac-collision -multihop-relay`, balanced n=3200 per cell) and is used for large-scale ranking claims. The secondary tier (stress and sensitivity blocks) is retained for patch-vs-control and tx-power diagnostics, not for replacing the primary ranking matrix.

Table 1. Evidence-regime map used in this manuscript.

Block	Scope	Typical sample size	Role in claims
100-node matrix	4 env × 5 protocols	n=30 per env-protocol cell	Core reliability comparison
Primary large-scale matrix	4 env × 6 node counts × 5 protocols	n=3200 per env-node-protocol cell	Primary large-scale evidence
Stress block A (patch-control)	4 env × 6 node counts × 5 protocols	patch n=1000, control n=600	Physics-stress calibration
Sensitivity block (tx-power)	4 env × 3 node counts × 5 protocols	n=600 per env-node-protocol cell	Tx-power boundary analysis
Stress block B (matched patch-control)	4 env × 6 node counts × 5 protocols	patch n=1000, control n=1000	Matched strict-physics confirmation
NS-3 alignment	4 env × 7 node counts, AERIS vs LEACH	n=30 per env-node-protocol cell	Trend-level cross-platform check

Table 1 defines the claim boundary used throughout the Results and Discussion sections.

4.5. Statistical Methods

For pairwise protocol comparisons, we use Welch's two-sample t-test:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}, \tag{6}$$

because variance equality is not assumed across protocols and environments. Family-wise error control is handled by Holm correction within each reported comparison family. Effect size is reported with Hedges' g:

$$g = J \cdot \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad J = 1 - \frac{3}{4(n_1 + n_2) - 9}, \tag{7}$$

and interpreted as small (0.2), medium (0.5), and large (0.8) in absolute value. Confidence intervals are reported as two-sided 95% intervals. All significance statements in this manuscript are based on Holm-corrected p values. For large-n cells, interpretation is based jointly on corrected p values, effect size, and absolute PDR delta. Reported table-wise standard deviations are treated as descriptive values from the corresponding result files; inferential conclusions are computed from raw replicate-level records.

4.6. Reproducibility Controls

All publication-tier scripts in this study use deterministic seed lists. They also enforce `force_ctp_reliable=False`. This guarantees that PDR reflects simulated delivery behavior rather than any reliability override. We report `pdr_expected` as the primary metric and treat hop count as a latency proxy, not a wall-clock latency measurement. Significance statements are accepted only when consistent with the declared regime-specific sample size.

4.7. Auditability Constraints

To keep claims machine-checkable, each publication-tier output includes run-tier tags, metric tags, and script/hash sidecar metadata. Claims in the main text are restricted to matrices that satisfy declared sample-size and metric constraints. This prevents mixing exploratory diagnostics with publication-tier statements.

5. Results

5.1. Multi-Environment Comparison at 100 Nodes (n=30)

Table 2. PDR by environment at 100 nodes from the legacy 100-node matrix (n=30 per environment-protocol cell; strict collision/relay flags disabled). Standard deviations are reported as stored in the source result file.

Environment	AERIS	LEACH	PEGASIS	HEED	TEEN
indoor_office	0.9739±0.0047	0.5543±0.0401	0.9078±0.0166	0.9371±0.0076	0.8222±0.0044
indoor_factory	0.6031±0.0258	0.1614±0.0209	0.1928±0.0255	0.2326±0.0263	0.3113±0.0245
outdoor_urban	0.3745±0.0354	0.0552±0.0127	0.0542±0.0117	0.0635±0.0121	0.1201±0.0183
outdoor_suburban	0.7451±0.0193	0.2703±0.0272	0.3382±0.0329	0.4221±0.0313	0.4752±0.0236

All pairwise AERIS-versus-baseline comparisons in Table 2 are significant after Holm correction ($p < 0.001$).

At 100 nodes, AERIS ranks first in all four tested environments within the evaluated baseline set (LEACH, PEGASIS, HEED, and TEEN). This claim is restricted to the 100-node matrix and should not be generalized to the 1000-node regime. Because Table 2 is generated from the legacy 100-node regime (without strict collision/relay flags), its absolute values are intentionally not pooled with the primary large-scale matrix.

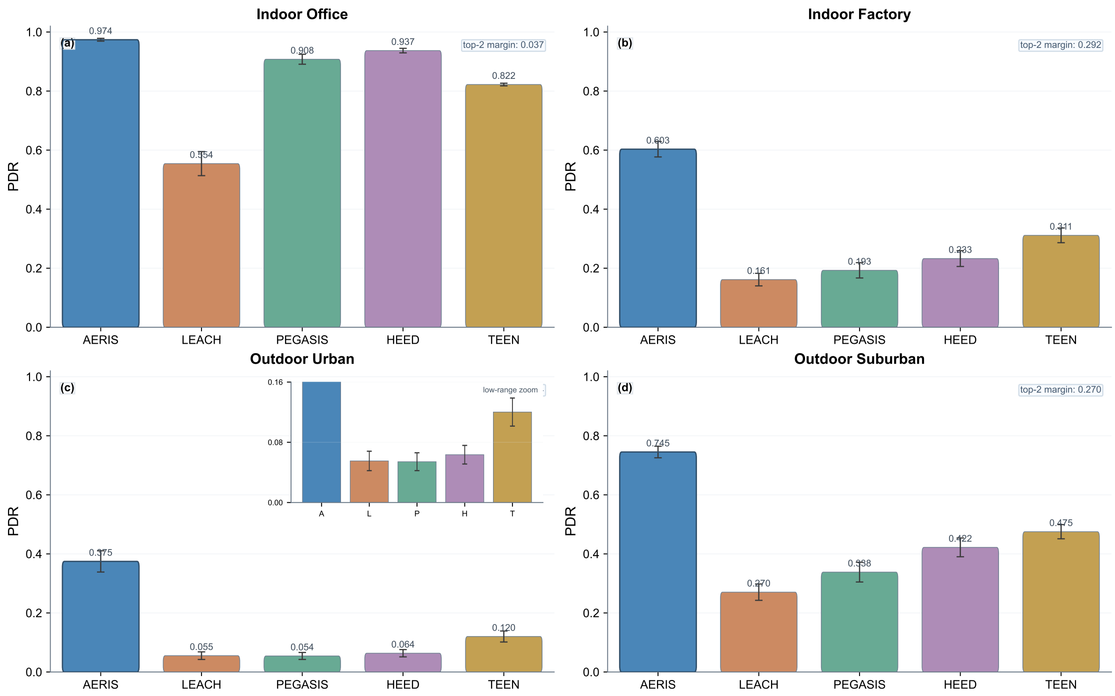

Figure 2. PDR comparison panel at 100 nodes (4 environments, n=30).

Figure 2 visually confirms the cross-environment ordering reported in Table 2.

5.2. Ablation: Environment-Dependent Effects (n=30)

Table 3. Full vs no_gateway PDR (mean ± std; standard deviations are reported as stored in the source result file).

Environment	Full	no_gateway	Delta (no_gateway - full)
indoor_office	0.9739±0.0047	0.9741±0.0036	+0.0002
indoor_factory	0.6031±0.0258	0.5806±0.0215	-0.0225
outdoor_urban	0.3745±0.0354	0.3534±0.0301	-0.0212
outdoor_suburban	0.7451±0.0193	0.7306±0.0264	-0.0146

Gateway is positive in three environments and near-neutral in indoor_office. CAS is not consistently positive across environments. In indoor_office, the Full-vs-no_gateway gap is +0.0002 (absolute PDR), so this cell is treated as practically neutral. These effects are quantified in Table 3.

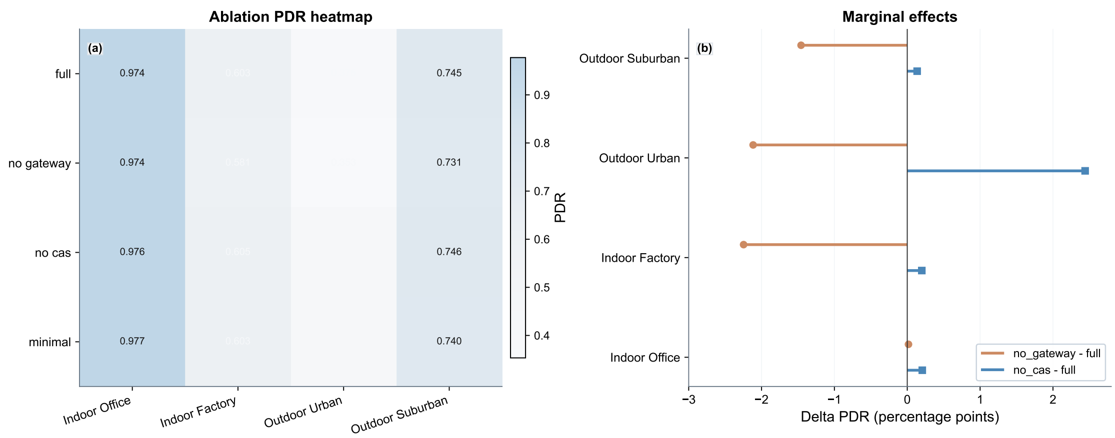

Figure 3. Ablation panel: heatmap and marginal effects (n=30).

Figure 3 highlights the environment dependence of gateway and CAS contributions.

5.3. Scalability (100–1000 Nodes; Balanced n=3200 per Cell)

Table 4. PDR ranking at 1000 nodes from the primary large-scale matrix (n=3200 per cell).

Environment	AERIS	LEACH	PEGASIS	HEED	TEEN	AERIS rank
indoor_office	0.6771	0.0282	0.9884	0.0134	0.0170	2nd
indoor_factory	0.7284	0.0108	0.2999	0.0054	0.0070	1st
outdoor_urban	0.1359	0.0041	0.0987	0.0016	0.0025	1st
outdoor_suburban	0.7275	0.0160	0.4725	0.0088	0.0106	1st

AERIS is first in three environments at 1000 nodes (indoor_factory, outdoor_urban, outdoor_suburban) and second in indoor_office where PEGASIS remains dominant (Table 4). Across the full 24-cell matrix (4 environments × 6 node counts), AERIS ranks first in 18 cells and PEGASIS ranks first in the 6 indoor_office cells. Unlike the legacy preliminary path, the primary large-scale matrix shows physically plausible scale behavior: AERIS PDR decreases from 100 to 1000 nodes in all four environments under explicit collision and relay constraints.

5.4. Statistical Robustness Snapshot (AERIS vs Strongest Baseline)

Table 5. Holm-corrected significance snapshot at 1000 nodes (AERIS vs strongest baseline in each environment).

Environment	Nodes	ΔPDR	Hedges' g	Holm p	Significant
indoor_office	1000	-0.311272 (vs PEGASIS)	-14.2041	$< 10^{-300}$	yes
indoor_factory	1000	+0.428522 (vs PEGASIS)	+5.0292	$< 10^{-300}$	yes
outdoor_urban	1000	+0.037265 (vs PEGASIS)	+0.5592	8.43×10^{-107}	yes
outdoor_suburban	1000	+0.255009 (vs PEGASIS)	+2.8235	$< 10^{-300}$	yes

This snapshot shows positive AERIS deltas in three harsh environments and a negative delta in indoor_office, where PEGASIS is the strongest baseline at large scale. The corresponding statistics are reported in Table 5. For very large sample sizes, practical interpretation in this study still follows both effect size and absolute PDR gap rather than significance labels alone. The very large absolute values of Hedges' g in harsh environments reflect both large mean gaps and low within-group variance under large sample sizes; they should be interpreted as magnitude indicators within this experiment design, not as cross-study constants.[1]

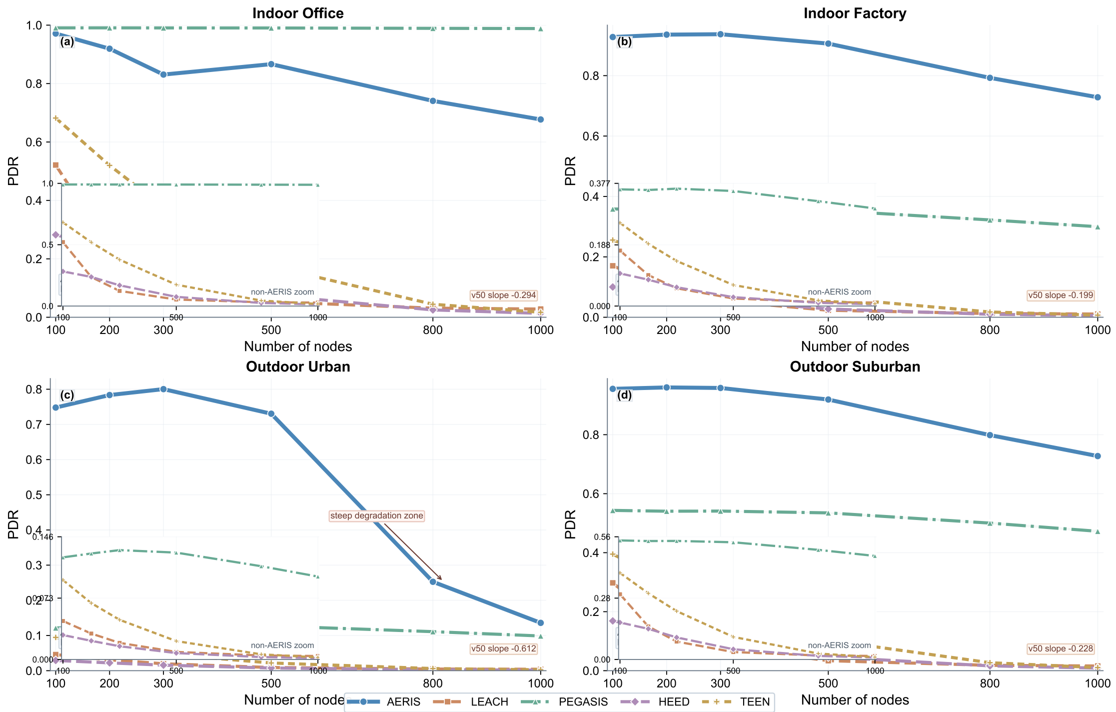

Figure 4. Primary large-scale scalability trends over node counts (balanced n=3200 per environment-node-protocol cell) with 95% CI bands. Insets show non-AERIS protocol details to avoid low-range compression. In several cells, CI bands are narrower than line width due to large sample size.

Figure 4 together with Table 5 provides the trend-and-significance view of the primary large-scale matrix.

[1] For large-n comparisons with very small within-group variance, standardized effect sizes can become extremely large. In this manuscript, Hedges' g is therefore interpreted jointly with absolute ΔPDR and confidence intervals, not as a stand-alone cross-study scale.

5.5. Rigor-Patch Pilot Sanity Check (n=60)

To test simulator-physics consistency before the full rerun, we ran a local pilot with collision and baseline multi-hop relay enabled (`-mac-collision -multihop-relay`), using the same environment taxonomy and nodes {100, 500, 1000} with n=60 per cell. This pilot is retained as a diagnostic snapshot and is not used in primary claim tables.

Table 6. AERIS PDR under the rigor-patch pilot (n=60 per environment-node cell).

Environment	100 nodes	500 nodes	1000 nodes
indoor_office	0.9706	0.8676	0.6746
indoor_factory	0.9283	0.9083	0.7355
outdoor_urban	0.7487	0.7362	0.1492
outdoor_suburban	0.9563	0.9193	0.7307

The key pilot observation is monotonicity recovery: AERIS PDR decreases with scale in all four environments. In indoor_office, PEGASIS is higher than AERIS under this stricter setting, while AERIS remains clearly higher in the other three environments. The pilot therefore served as a gate check for launching the full primary large-scale matrix and is reported only for diagnostic completeness. The pilot values summarized above are reported in Table 6.

5.6. Patch-vs-Control Stress Comparison (Four Environments)

To reduce seed and sampling confounders, we ran a matched patch-versus-control stress test in all four environments (indoor_office, indoor_factory, outdoor_urban, and outdoor_suburban). The patch mode enables collision and baseline relay mechanisms; the control mode uses the original simulator path. Patch cells use n=1000 and control cells use n=600.

Table 7. AERIS PDR in the patch-vs-control stress comparison (four environments).

Environment	Nodes	Patch	Control
indoor_office	100	0.9714	0.9948
indoor_office	500	0.8679	0.9902
indoor_office	1000	0.6779	0.9899
indoor_factory	100	0.9281	0.9329
indoor_factory	500	0.9060	0.9627
indoor_factory	1000	0.7291	0.9726
outdoor_urban	100	0.7482	0.7559
outdoor_urban	500	0.7299	0.8460
outdoor_urban	1000	0.1372	0.8845
outdoor_suburban	100	0.9556	0.9744
outdoor_suburban	500	0.9198	0.9859
outdoor_suburban	1000	0.7280	0.9897

Table 7 provides the patch-vs-control stress baseline used for cross-regime interpretation.

Three observations are robust in this stress comparison matrix. First, AERIS remains above LEACH/HEED/TEEN under patch mode in all four environments at 1000 nodes. Second, PEGASIS is minimally affected by the patch, with near-zero deltas and non-significant tests in all 24 PEGASIS cells[2]. Third, AERIS patch-control deltas are significant in all 24 AERIS cells after Holm correction, with the largest absolute drop in outdoor_urban at high scale. Table 8 provides the 1000-node PEGASIS snapshot supporting this interpretation.

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[2] Across the 24 PEGASIS cells in this stress comparison matrix, $|\Delta_{\text{patch-control}}| \leq 0.0044$, all Holm-adjusted p-values are 1.0000, and max $|g| = 0.0903$, consistent with a near-zero effect.

In the matched patch-control confirmation matrix, PEGASIS shows exact zero deltas in indoor_factory (all six node counts). We treat this as an implementation-coupling anomaly pending code-path audit, not as physical invariance evidence. The patch can also be overly aggressive for baseline paths at high scale, so this block is interpreted as stress evidence rather than as a final replacement matrix. To remove the sample-size asymmetry in the stress comparison matrix, we run a matched patch-control confirmation matrix (patch $n = 1000$, control $n = 1000$ per cell) and report it separately below.

Consistency between the stress comparison matrix and the matched confirmation matrix is assessed from the matched-delta tables and significance files; both matrices preserve the same direction pattern for AERIS across environment-scale cells.

Table 8. PEGASIS patch-versus-control snapshot at 1000 nodes from the stress comparison matrix (n=1000 patch, n=600 control).

Environment	Delta (patch-control)	Hedges' g	Holm p	Significant
indoor_office	-0.0003	-0.0435	1.0000	No
indoor_factory	+0.0006	+0.0057	1.0000	No
outdoor_urban	+0.0017	+0.0252	1.0000	No
outdoor_suburban	-0.0011	-0.0085	1.0000	No

5.7. TX-Power Sensitivity Block (Four Environments)

We additionally evaluate tx-power sensitivity under the upgraded simulation path by comparing 5 dBm versus 15 dBm across four environments, three node counts (100, 500, 1000), and five protocols (n=600 per cell). The full matrix contains 60 matched comparisons (4 environments $\times$ 3 node counts $\times$ 5 protocols), of which 59/60 are significant after Holm correction.

Table 9. AERIS power-sensitivity snapshot at 1000 nodes (mean PDR).

Environment	tx=5 dBm	tx=15 dBm	Delta (5–15)
indoor_office	0.8177	0.5367	+0.2810
indoor_factory	0.6623	0.5051	+0.1571
outdoor_urban	0.0587	0.1606	-0.1019
outdoor_suburban	0.7679	0.5440	+0.2239

Table 9 reports the 1000-node AERIS tx-power snapshot used in this subsection.

Figure 5. Full-matrix power sensitivity (tx5 vs tx15). Panels (a)–(d) show per-environment protocol-by-scale delta maps (ΔPDR = tx5 − tx15) for all 60 tested cells; cross markers denote non-significant cells after Holm correction. Panel (e) summarizes protocol-level absolute sensitivity across 4 environments $\times$ 3 node scales.

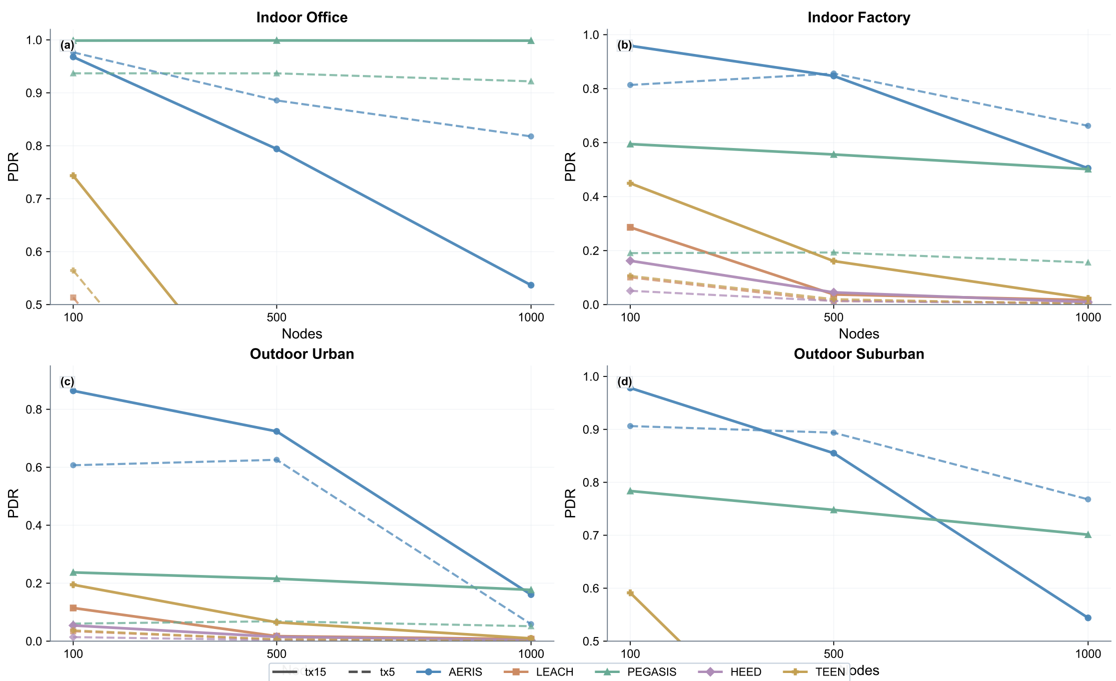

Figure 6. Absolute PDR profiles under tx5 (dashed) and tx15 (solid) across the same 60-cell matrix. This view complements Figure 5 by showing absolute levels rather than only deltas.

The tx-power block confirms that response is environment-dependent and non-monotonic under the current routing and channel settings. In particular, raising tx-power does not guarantee higher PDR in every environment-node cell. A plausible mechanism is interference-domain expansion: higher transmit power increases both reachable receivers and competing transmitters, which can increase collision probability in dense regimes and partially offset link-budget gains. The single non-significant comparison in the 60-cell matrix is LEACH at indoor_office, 1000 nodes; all other 59 cells are significant after Holm correction. Figures 5 and 6 are interpreted together with Table 9 to avoid delta-only bias.

5.8. Matched Patch-vs-Control Confirmation Matrix (Four Environments, n=1000 vs 1000)

To remove remaining sample-size asymmetry from the stress comparison matrix, we completed a fully matched patch-vs-control confirmation matrix in which both patch and control arms use n=1000 per environment-node-protocol cell (4 environments $\times$ 6 node counts $\times$ 5 protocols). This matrix directly tests whether stricter physics assumptions improve or degrade reliability under matched sampling.

Table 10. AERIS patch-control deltas in the matched confirmation matrix (delta = patch - control).

Environment	Delta at 100 nodes	Delta at 1000 nodes
indoor_office	-0.0234	-0.3120
indoor_factory	-0.0052	-0.2435
outdoor_urban	-0.0077	-0.7474
outdoor_suburban	-0.0191	-0.2616

Table 10 summarizes the AERIS matched-delta boundary in the confirmation matrix.

In the matched matrix, AERIS patch-control deltas are negative and significant in all 24 environment-scale cells after Holm correction (24/24 significant; Hedges' g range: -0.47 to -15.36). Therefore, under the current patch configuration (`mac-collision + multihop-relay`), stricter physics assumptions reduce reliability rather than increase it. This means the patch block is evidence for realism stress, not yet evidence for superior calibrated performance.

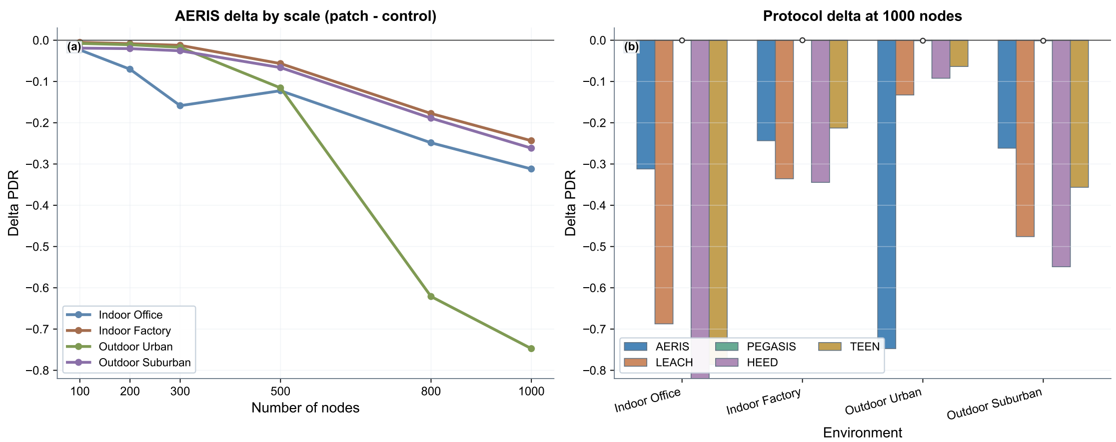

Figure 7. Matched patch-control evidence. (a) AERIS delta trajectories over node counts (patch - control). (b) Protocol deltas at 1000 nodes across environments. Negative deltas indicate lower PDR under patch mode.

At the protocol level, Holm-significant cell counts in the matched patch-control block are 24/24 for AERIS, 24/24 for LEACH, 0/24 for PEGASIS, 24/24 for HEED, and 24/24 for TEEN. Together with Figure 7, this shows that the strict patch strongly affects most protocol families, while PEGASIS remains nearly invariant in this block.

5.9. Hop-Based Latency and Trade-offs

At 100 nodes (n=30), AERIS stays near two hops (1.97–1.99), while PEGASIS remains above 31 hops due to chain traversal. LEACH and TEEN show lower average hops because of protocol paths that allow direct-to-BS transmissions.

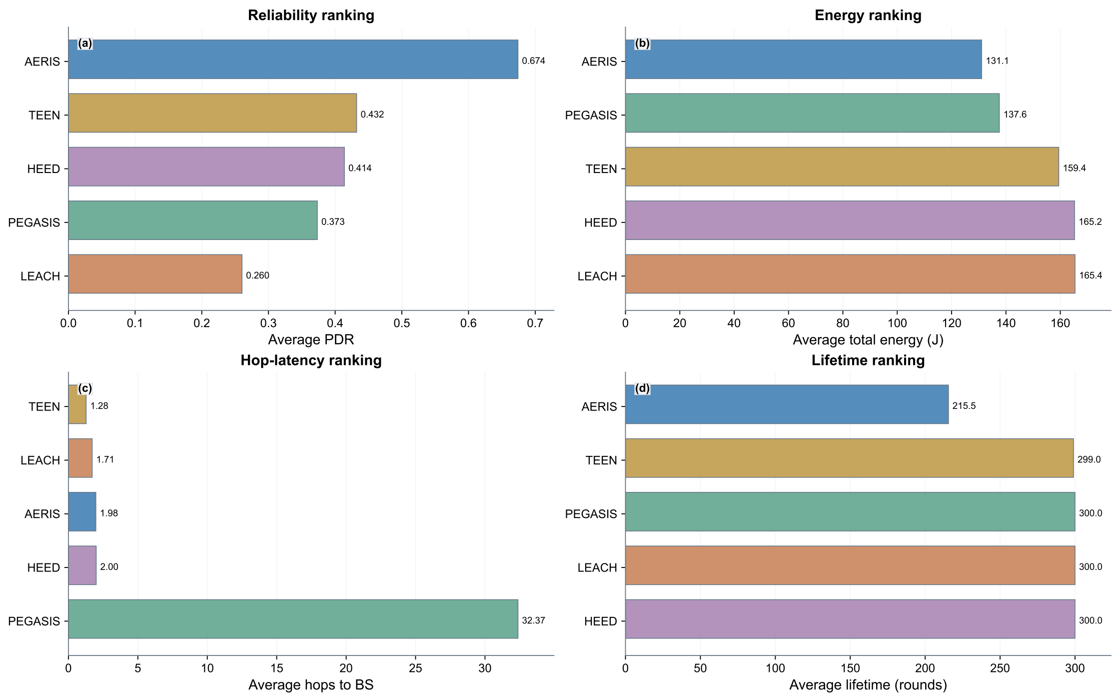

Figure 8. Trade-off panel from publication-tier data: reliability, energy, hop-based latency, and lifetime. The hop panel uses a logarithmic x-axis to keep low-hop and high-hop protocols visible in the same frame without overlap.

Figure 8 is used as the consolidated reliability-energy-latency-lifetime trade-off view.

5.10. NS-3 Trend Validation (AERIS vs LEACH, n=30)

NS-3 validation is used as a cross-platform trend check rather than a numerical-equivalence proof. This block is restricted to AERIS-versus-LEACH and is not used for five-protocol cross-platform ranking. Under aligned multi-environment settings, AERIS is directionally not lower than LEACH in 27 of 28 tested environment-scale cells; the single exception is indoor_office at 200 nodes (AERIS–LEACH diff $= -0.0004$, not significant). At 100 nodes, significance is confirmed in 3/4 environments (indoor_factory, outdoor_urban, outdoor_suburban), while indoor_office remains non-significant. In the 300/500-node extension, all eight environment-scale comparisons are significant after Holm correction; with the 800/1000 extension included, non-significance remains confined to indoor_office at 1000 nodes.

Table 11. NS-3 trend-level validation at 100 nodes (AERIS vs LEACH, n=30; metric: mean PDR).

Environment	AERIS	LEACH	Delta	Holm-corrected p	Significance
indoor_office	0.9202	0.9185	+0.0017	1.00	no
indoor_factory	0.6025	0.5336	+0.0690	4.23×10^{-25}	yes
outdoor_urban	0.2064	0.1899	+0.0165	3.41×10^{-3}	yes
outdoor_suburban	0.7771	0.6921	+0.0850	1.94×10^{-30}	yes

Table 11 provides the 100-node NS-3 trend snapshot referenced in this subsection.

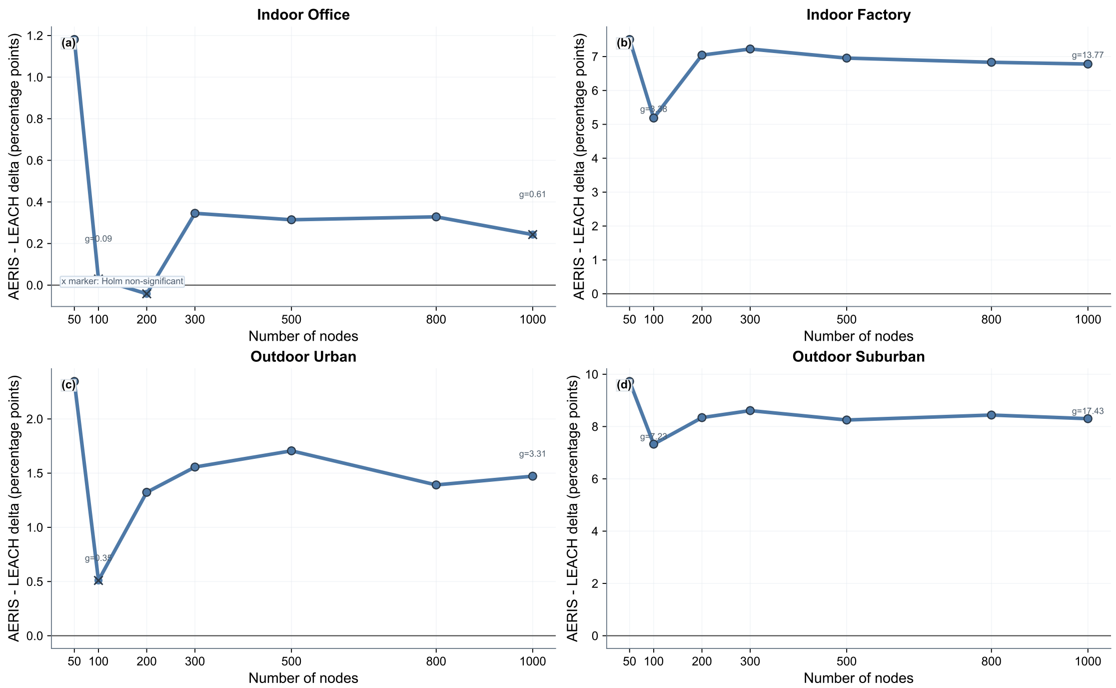

Figure 9. NS-3 trend panel over 50–1000 nodes (AERIS–LEACH delta in percentage points, pp). Cross markers denote non-significant cells after Holm correction.

Scope note: this NS-3 block supports trend consistency only. The manuscript does not claim cross-platform numerical equivalence. Across 7 node counts and 4 environments, 25/28 AERIS-versus-LEACH comparisons are significant; the non-significant cells are indoor_office at node counts 100, 200, and 1000. Figure 9 visualizes the full 50–1000 node NS-3 trend envelope.

6. Discussion

The evidence supports environment-scoped positioning rather than universal superiority. In the reported matrices, AERIS is strongest on reliability in harsh channels.

The manuscript separates three evidence blocks:

- 100-node multi-environment comparison,
- primary large-scale matrix (100–1000 nodes) with balanced n=3200 per cell,
- NS-3 trend validation for AERIS versus LEACH only.

Cross-block numerical pooling is avoided because these blocks answer different questions. Under stricter patch-control stress settings, high-scale PDR decreases for most protocols and PEGASIS remains comparatively stable. Under the tx-power sensitivity block, power effects are strong but not globally monotonic. Practical interpretation therefore remains environment-scoped.

Compared with recent adaptive-routing studies [12,13,15,16,21,22,26,30], this work contributes a reproducible large-sample evaluation envelope with explicit regime separation and cross-platform trend checks. The main distinction is methodological: rather than reporting a single favorable matrix, we report both primary ranking evidence and matched stress/sensitivity blocks that expose when absolute reliability degrades.

Practical interpretation also requires separating statistical and engineering significance. At large sample sizes, very small absolute differences can be significant after correction. For this reason, we interpret each result jointly by corrected p value, effect size, and absolute PDR delta. Energy interpretation is similarly conditioned: lower total energy can co-occur with shorter lifetime when earlier network termination occurs.

Interpretation of Matched Degradation Block

The matched patch-vs-control confirmation matrix shows that enabling the current patch configuration (MAC-collision plus multihop relay) reduces absolute AERIS PDR in all 24 matched cells. This does not contradict the primary ranking result, because ranking and absolute-level behavior answer different questions: (i) ranking asks whether AERIS remains higher than alternatives under the same strict setting, while (ii) the matched confirmation matrix asks how far the strict patch shifts each protocol relative to its own control path. In our data, strict-physics penalties affect all protocols, but the penalty profile is not uniform across protocol families. Therefore, the correct interpretation is not "AERIS improves under patch", but "AERIS keeps relative advantages in harsh regimes while absolute reliability drops under stricter channel/relay assumptions." This distinction is central to deployment calibration.

Practical Deployment Guidance

Based on the completed matrices, the following deployment guidance is supported by the reported evidence:

- In harsh environments (indoor_factory, outdoor_urban, outdoor_suburban), enabling Gateway support is recommended because no-gateway ablations reduce reliability.
- In benign indoor_office conditions, gains are smaller and model overhead should be justified against application reliability targets.
- For high-scale deployments, use the primary large-scale matrix as primary evidence and use stress/sensitivity blocks for calibration.
- Tx-power settings should be environment-tuned; monotonic power assumptions are not supported by the sensitivity block.

Table 12. Environment-scoped engineering interpretation from the reported evidence.

Environment type	Recommended interpretation	Main caveat
indoor_office	Reliability gaps are small at high scale; deployment cost and simplicity can dominate design choice	Some NS-3 cells are non-significant
indoor_factory	AERIS with Gateway support gives the most stable reliability gain	Calibrate tx-power and collision parameters
outdoor_urban	AERIS remains preferable under harsh links in both simulator regimes	High-scale behavior is sensitive to physics settings
outdoor_suburban	AERIS trend remains favorable in simulator and NS-3 trend checks	Do not assume monotonic power-response

Table 12 summarizes the actionable interpretation boundaries for deployment decisions.

Validity Notes

Three boundaries are explicit in this draft. First, the reported comparisons are simulator-based under the tested channel taxonomy. Second, NS-3 evidence is used for cross-platform trend validation (AERIS vs LEACH), not full five-protocol numerical alignment. Third, module contributions are context dependent: Gateway and CAS effects vary by environment and are not globally monotonic.

7. Limitations and Future Work

- Current evidence is simulation-based under the tested channel taxonomy.
- NS-3 validation in this manuscript focuses on AERIS versus LEACH; full five-protocol cross-platform ranking is future work.
- The primary large-scale matrix (n=3200 per cell) includes explicit collision and relay constraints, but does not replace hardware-in-the-loop validation.

- The patch-vs-control stress matrix, the tx-power sensitivity matrix, and the matched patch-control confirmation matrix show that stricter physics assumptions and tx-power changes can materially alter absolute PDR, so deployment requires environment-specific calibration.
- In the matched patch-control confirmation matrix, PEGASIS shows near-invariant patch-control deltas; this behavior is treated as a simulator-coupling artifact pending dedicated code-path audit.
- Reported gains are limited to the tested baseline set and topology families in this manuscript.
- Hop count is a route-length proxy, not a wall-clock latency benchmark under full MAC scheduling.

8. Conclusion

AERIS is a reliability-oriented routing option for heterogeneous WSN channels. In the publication-tier 100-node matrix, it achieves the highest mean PDR within the tested baseline set across all four environments. In the primary large-scale matrix (100–1000 nodes, n=3200 per cell, explicit collision/relay modeling), AERIS is rank-1 in three environments (indoor_factory, outdoor_urban, outdoor_suburban) and rank-2 in indoor_office where PEGASIS remains dominant.

NS-3 provides trend-level cross-platform support for AERIS versus LEACH. The completed stress and sensitivity blocks show that stricter physics assumptions and tx-power changes materially affect high-scale behavior. In the matched patch-control confirmation matrix, AERIS patch-control deltas are negative and significant in all 24 environment-scale cells under the current patch setup.

The practical conclusion is environment-scoped deployment: AERIS remains strong for reliability-first operation in harsh channels, while benign indoor conditions at large scale may favor chain-forwarding baselines. Final parameterization should therefore be environment-tuned using matched stress and sensitivity matrices.

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Data Availability Statement

The datasets, scripts, manuscript sources, figure-generation code, and provenance sidecar records (including commit identifiers and script hashes) supporting this study are available from the corresponding author and will be released in a versioned public repository upon final publication. The released archive includes the full 100-node benchmark matrix, the primary large-scale matrix and significance tables, stress/sensitivity calibration matrices, and the NS-3 trend-level validation summary.

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